

**A Major Project Report**

**On**

**“DEEP FAKE AUDIO DETECTION USING DEEP LEARNING”**

Submitted in Partial Fulfilment of the Academic Requirement for the Award of Degree  
of

**BACHELOR OF TECHNOLOGY**

**In**

**DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING**

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Under the Esteemed Guidance of

**Mrs. K. Priyanka**

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**CMR INSTITUTE OF TECHNOLOGY**

**(UGC AUTONOMOUS)**

Approved by AICTE, Affiliated to JNTUH, Accredited by NAAC with 'A+' Grade

Kandlakoya(V), Medchal Dist-501 401

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**2025-26**

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DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING



## CERTIFICATE

This is to certify that a Major Project entitled with “**Deep fake audio detection using deep learning**” is being submitted by:

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To JNTUH, Hyderabad, in partial fulfillment of the requirement for award of the degree of **B.Tech** in **Computer Science and Engineering** and is a record of a bonafide work carried out under our guidance and supervision. The results in this project have been verified and are found to be satisfactory. The results embodied in this work have not been submitted to have any other University for award of any other degree or diploma.

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## ACKNOWLEDGEMENT

We are extremely grateful to **Dr. M Janga Reddy**, Director, **Dr. B. Satyanarayana**, Principal and **Dr. K. Pradeep Reddy**, Head of Department, Department of Computer Science and Engineering, CMR Institute of Technology for their inspiration and valuable guidance during entire duration.

We are extremely thankful to **Mrs. L. Divya**, Assistant Professor, Major Project Coordinator and internal guide “**Mrs. K. Priyanka**”, Assistant Professor Dept. of Computer Science and Engineering, CMR Institute of Technology for their constant guidance, encouragement and moral support throughout the project.

We will be failing in duty if we do not acknowledge with grateful thanks to the authors of the references and other literatures referred in this Project.

We express our thanks to all staff members and friends for all the help and coordination extended in bringing out this Project successfully in time.

Finally, we are very much thankful to our parents and relatives who guided directly or indirectly for every step towards success.

|                         |                   |
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## **ABSTRACT**

With the proliferation of deep learning-based voice synthesis technologies, deep fake audio has become an emerging cybersecurity and ethical threat. This project proposes a deep learning-based system for the detection of deep fake audio using advanced neural network architectures. The model leverages spectrogram analysis and convolutional neural networks (CNNs) to learn distinguishing features between authentic and synthesized audio. A dataset consisting of both real and AI-generated audio samples was compiled, preprocessed, and used to train and evaluate the model. Feature extraction techniques, such as Mel-frequency cepstral coefficients (MFCC) and log-mel spectrograms, were employed to convert audio into image-like representations for input into the CNN. Experimental results indicate that the model achieves high classification accuracy in detecting deep fake audio across various speakers and synthetic voice generation techniques.

This research demonstrates the potential of deep learning in combating the challenges posed by deep fake technologies. Future work may involve incorporating recurrent neural networks (RNNs) for better temporal feature analysis, as well as expanding the model's robustness across languages and background noise conditions..

## TABLE OF CONTENTS

|                                      |              |
|--------------------------------------|--------------|
| <b>1. Introduction</b>               | <b>1</b>     |
| <b>2. Literature Survey</b>          | <b>2-3</b>   |
| <b>3. System Analysis</b>            | <b>4-6</b>   |
| 3.1 Existing System                  | 4            |
| 3.2 Disadvantages of Existing System | 4            |
| 3.3 Proposed System                  | 5            |
| 3.4 Advantages of Proposed System    | 5-6          |
| <b>4. System Study</b>               | <b>7-8</b>   |
| 4.1 Feasibility Study                | 7-8          |
| 4.1.1 Technical Feasibility          | 7            |
| 4.1.2 Economical Feasibility         | 7            |
| 4.1.3 Operational Feasibility        | 7            |
| 4.1.4 Social feasibility             | 8            |
| <b>5. System Requirements</b>        | <b>9</b>     |
| 5.1 Hardware Requirements            | 9            |
| 5.2 Software Requirements            | 9            |
| <b>6. Objective Of The Project</b>   | <b>10</b>    |
| <b>7. Architecture</b>               | <b>11</b>    |
| <b>8. Modules</b>                    | <b>12-13</b> |
| <b>9. Diagrams</b>                   | <b>14-17</b> |
| 9.1 Data flow Diagrams               | 14           |
| 9.2 Use case diagrams                | 15           |
| 9.3 Sequence Diagrams                | 16           |
| 9.4 Collaborative Diagram            | 17           |
| <b>10.Implementation</b>             | <b>18-43</b> |
| 10.1 Source Code                     | 18-43        |
| <b>11.Screen Shots</b>               | <b>44-51</b> |
| <b>12.Testing</b>                    | <b>52-54</b> |
| <b>13.Conclusion</b>                 | <b>55</b>    |
| <b>14.References</b>                 | <b>56</b>    |

## LIST OF FIGURES

| Figure No. | Particulars           | Page No. |
|------------|-----------------------|----------|
| 7.1        | Architecture          | 11       |
| 9.1        | Data Flow Diagram     | 14       |
| 9.2        | Use Case Diagram      | 15       |
| 9.3        | Sequence Diagram      | 16       |
| 9.4        | Collaborative Diagram | 17       |

## 1.INTRODUCTION

Deepfake audio refers to synthetic audio clips generated using machine learning models that closely mimic a person's voice, speech patterns, and emotional tone. These models can create audio that is often indistinguishable from real recordings, leading to critical concerns in areas such as media integrity, legal evidence, telecommunications, and cybersecurity.

With the advent of generative models like GANs (Generative Adversarial Networks), autoencoders, and transformer-based architectures, the creation of high-quality deepfake audio has become more accessible. Consequently, the development of effective detection mechanisms has become a vital area of research.

This project explores the use of **deep learning techniques** for detecting deepfake audio. It involves analyzing the spectrograms, waveforms, and acoustic features of audio clips using convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid models to classify audio as genuine or synthetic. By leveraging large annotated datasets and advanced model architectures, the system aims to achieve high accuracy in detecting forgeries and contribute toward safer digital communication environments.

## **2.LITERATURE SURVEY**

### **2.1 Understanding the Evolution of Deepfake Technologies**

Deepfake audio generation has improved significantly with the development of advanced models such as WaveNet, Tacotron, and GAN-based speech synthesis. According to Westerlund (2019), deepfake technology has evolved to produce highly realistic audio that closely mimics human speech. These models can generate natural voice patterns, making it difficult to distinguish between real and fake audio.

### **2.2 Identifying Existing Detection Approaches**

Several research works have proposed methods for detecting deepfake audio. Rössler et al. (2019) used deep learning techniques for detecting manipulated media, while Koay et al. (2021) applied Convolutional Neural Networks (CNNs) on spectrograms for audio classification. Other approaches include the use of Recurrent Neural Networks (RNNs) and attention mechanisms to capture temporal patterns in speech.

However, many of these methods face challenges in detecting high-quality deepfakes and often depend on specific datasets.

### **2.3 Analyzing Available Datasets and Benchmarks**

Public datasets such as ASVspoof (Kinnunen et al., 2019), FakeAVCeleb, and WaveFake are widely used for training and evaluation. These datasets contain both real and synthetic audio samples. The ASVspoof dataset, in particular, is considered a benchmark for evaluating spoofing detection systems.

Using diverse datasets improves the robustness and generalization ability of detection models.

### **2.4 Studying Feature Extraction Techniques**

Feature extraction plays a crucial role in deepfake detection. Techniques such as Mel-Frequency Cepstral Coefficients (MFCC) and spectrogram analysis are commonly used. Tak et al. (2021) demonstrated that analyzing raw waveform and spectral features can improve detection accuracy.



These features help in identifying subtle differences in frequency and speech patterns between real and fake audio.

### 2.5 Evaluating Metrics and Model Performance

Different evaluation metrics are used to measure the performance of deepfake detection systems. Common metrics include accuracy, Equal Error Rate (EER), Area Under Curve (AUC), and F1-score. Studies show that deep learning models, especially hybrid models like CNN + LSTM, achieve higher accuracy compared to traditional methods.

However, performance may vary depending on dataset quality and model complexity.

### 3.SYSTEM ANALYSIS

#### 3.1.EXISTING SYSTEM:

Most existing deepfake detection methods focus on **video deepfakes**, leaving audio detection relatively underexplored. The few available systems often rely on **handcrafted acoustic features**, such as pitch, tempo, and MFCCs (Mel-Frequency Cepstral Coefficients), or on traditional machine learning classifiers like SVM or decision trees. However, these methods are **limited in accuracy**, especially against sophisticated, GAN-based audio synthesis models that mimic human vocal nuances extremely well.

Additionally, existing models often suffer from:

- **Low generalizability** to new or unseen deepfake generation techniques.
- **Inability to adapt** to various languages, accents, and background noises.
- **Lack of real-time detection capabilities**, making them unsuitable for live applications like call monitoring.

These limitations highlight the inadequacy of conventional methods in combating the evolving landscape of deepfake audio threats.

#### 3.2.DISADVANTAGES OF EXISTING SYSTEMS:

1. **Low accuracy** with high-quality synthesized audio, especially from advanced TTS systems.
2. **High false positives/negatives** due to inability to adapt to diverse voice patterns.
3. **Lack of generalization** across different languages, accents, and audio environments.
4. **Manual feature extraction** limits scalability and fails to capture subtle deepfake cues.
5. **Delayed detection** and inefficiency in real-time applications.

### 3.3.PROPOSED SYSTEM:

To address the shortcomings of current systems, this project proposes a **Deep Learning-based Deep Fake Audio Detection System**. The proposed system leverages **neural networks trained on large datasets of real and synthetic speech** to learn the subtle patterns and anomalies that distinguish fake audio from genuine recordings.

Key components of the proposed system include:

- **Spectrogram-Based Feature Extraction:** Converting audio signals into spectrograms to capture frequency and temporal patterns, which are more effective for neural network training.
- **Convolutional Neural Networks (CNNs):** Used to identify local patterns in spectrograms that are indicative of synthesis artifacts.
- **Recurrent Neural Networks (RNNs) or LSTM models:** Employed to analyze temporal dependencies in speech, enabling the detection of unnatural transitions or pacing common in fake audio.
- **Real-Time Detection Capability:** Optimized model for low-latency inference to support live audio analysis in applications like telecommunication or virtual meetings.
- **Generalization Across Models:** Training the system with audio generated by various TTS and voice cloning models (e.g., Tacotron, WaveNet, DeepVoice) to improve robustness.

This deep learning approach provides a **more adaptive, accurate, and scalable solution** compared to conventional methods, making it a critical tool in the defense against deepfake audio threats.

### 3.4.ADVANTAGES OF THE PROPOSED SYSTEM:

- **Automatic Feature Extraction:**
  - Deep learning models can learn rich and high-dimensional features directly from raw or spectrogram representations of audio, eliminating the need for manual feature engineering.
- **Improved Accuracy:**

## Deep Fake Audio Detection using Deep Learning

- Models like CNNs and LSTMs can capture both spatial and temporal patterns, allowing for detection of subtle irregularities in audio signals caused by synthesis.
- **Generalization Capability:**
  - Trained on large and diverse datasets, the system can detect deepfakes across different speakers, languages, and synthesis techniques.

## 4. SYSTEM STUDY

The project is designed to detect deepfake audio using deep learning techniques. The system takes an audio file as input, processes it, extracts features like MFCC and spectrograms, and uses a trained CNN + LSTM model to classify whether the audio is real or fake.

The study mainly focuses on building an efficient and accurate system that can handle different types of audio and detect subtle differences between real and AI-generated voices.

### 4.1.FEASIBILITY STUDY

#### 4.1.1.Technical Feasibility:

The project is technically feasible as it uses widely available technologies such as Python and deep learning frameworks. The implementation of CNN and LSTM models is supported by libraries like TensorFlow and Keras. The system can be developed and executed on standard computers without requiring high-end hardware. The dataset required for training is also easily available from online sources.

#### 4.1.2. Economic Feasibility:

The project is cost-effective because it uses open-source tools and software. No paid licenses or expensive infrastructure is required. All development tools such as Python, TensorFlow, and libraries are freely available, making the project affordable and suitable for academic purposes.

#### 4.1.3. Operational Feasibility:

The system is user-friendly and easy to operate. Users can upload audio files through a simple interface and get results without requiring deep technical knowledge. The process of loading data, training the model, and detecting deepfake audio is straightforward, making the system practical for real-world use.

### **4.1.4. Social Feasibility:**

This project is socially beneficial as it helps in detecting fake audio, which can prevent misuse such as fraud, misinformation, and identity impersonation. It contributes to improving trust in digital communication and can be useful in areas like media verification, cybersecurity, and law enforcement.

## 5.SYSTEM REQUIREMENTS

### 5.1.HARDWARE REQUIREMENTS:

| Requirements | Features          |
|--------------|-------------------|
| System       | Pentium IV 2.4GHz |
| Hard Disk    | 40 GB             |
| Ram          | 512 Mb            |

### 5.2.SOFTWARE REQUIREMENTS:

| Requirements     | Features |
|------------------|----------|
| Operating System | Windows  |
| Language         | Python   |

## 6.OBJECTIVE OF THE PROJECT

- To design and develop a system for detecting deepfake audio using deep learning techniques
- To analyze audio signals and extract relevant features such as Mel-Frequency Cepstral Coefficients (MFCC) and spectrograms
- To implement a hybrid model using Convolutional Neural Network (CNN) and Long Short-Term Memory (LSTM) for accurate classification
- To identify and differentiate between real and AI-generated audio based on learned patterns
- To improve the overall accuracy and reliability of deepfake audio detection
- To provide a user-friendly interface for uploading audio and obtaining classification results



## 7.ARCHITECTURE

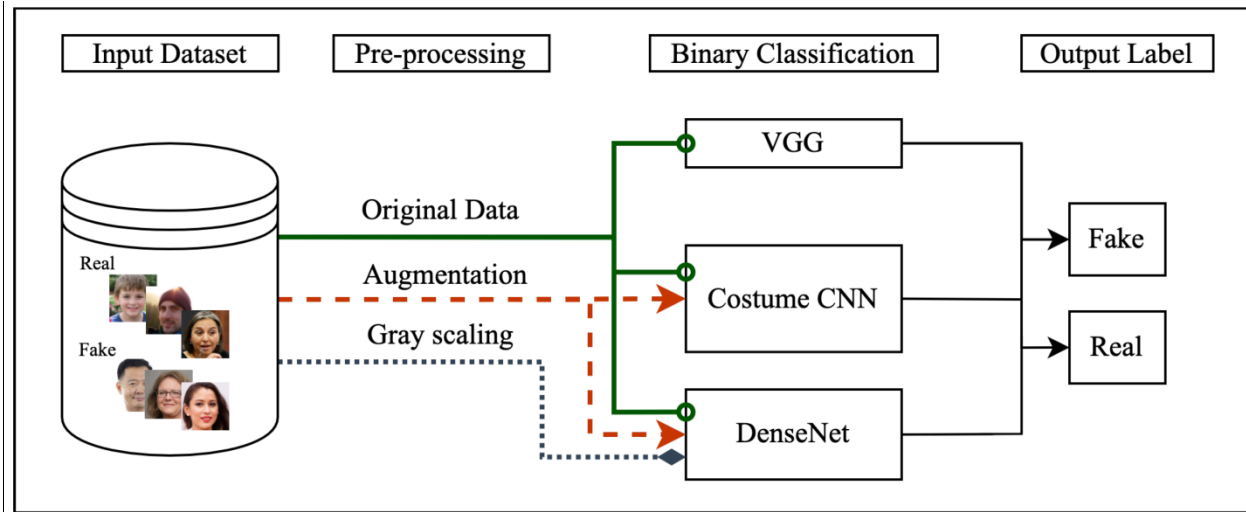


Fig.7.1

The system follows a pipeline approach to detect deepfake audio. It processes the input audio step by step and classifies it as real or fake.

### Architecture Flow:

- Audio Input: User uploads audio file
- Preprocessing: Audio is cleaned and normalized
- Feature Extraction: MFCC and spectrogram features are extracted
- Model (CNN + LSTM):
  - CNN extracts patterns
  - LSTM analyzes time sequence
- Classification: Audio is classified as real or fake
- Output: Result is displayed to the user

## 8. MODULES

### 8.1. DATA COLLECTION AND PREPROCESSING:

- **Dataset Sources:** Publicly available datasets such as ASVspoof, FakeAVCeleb, and custom-collected real and synthetic audio samples.
- **Feature Extraction:** Conversion of audio signals to spectrograms, MFCCs (Mel Frequency Cepstral Coefficients), or raw waveform inputs depending on the model architecture.
- **Normalization:** Ensuring consistent amplitude, length, and frequency ranges to standardize inputs across diverse sources.

### 8.2. MODEL ARCHITECTURE:

- **CNN-based Models:** Convolutional Neural Networks for learning local spectral-temporal features from spectrogram images.
- **RNN/LSTM Models:** Recurrent architectures that analyze temporal sequences in audio to detect subtle inconsistencies in speech patterns or prosody.
- **Transformer Models:** Attention-based deep learning models for capturing long-term dependencies and context-aware features in speech.
- **Ensemble Approach:** Combining multiple models to improve overall detection performance and reduce false positives/negatives.

### 8.3. TRAINING AND VALIDATION:

- **Training:** The model is trained on labeled datasets of real and fake audio samples using supervised learning.
- **Validation:** Performance metrics such as accuracy, precision, recall, F1-score, and ROC-AUC are used to evaluate the model.
- **Augmentation:** Data augmentation techniques such as pitch shift, speed change, and background noise addition are used to increase robustness.

#### 8.4. DETECTION AND INFERENCE ENGINE:

- **Real-time Detection:** The system is capable of processing input audio and outputting a classification in real time or near-real time.
- **Confidence Scoring:** Each prediction is accompanied by a confidence score indicating the likelihood of the audio being fake.

#### 8.5. USER INTERFACE AND ALERT SYSTEM:

- **Dashboard:** Visual interface for uploading audio, displaying results, and showing detection history.
- **Alert Mechanism:** Email or log-based alerts for flagged deepfake audio events in critical applications like media verification or law enforcement.

#### 8.6. DEPLOYMENT AND INTEGRATION:

- **APIs and SDKs:** The system can be packaged as a REST API or embedded in voice authentication systems.
- **Scalability:** Designed to be cloud-deployable with horizontal scalability for high-volume use.

## 9.DIAGRAMS

### 9.1. DATAFLOW DIAGRAM:

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a" or "has-a" relationship. Each class in the class diagram may be capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely.

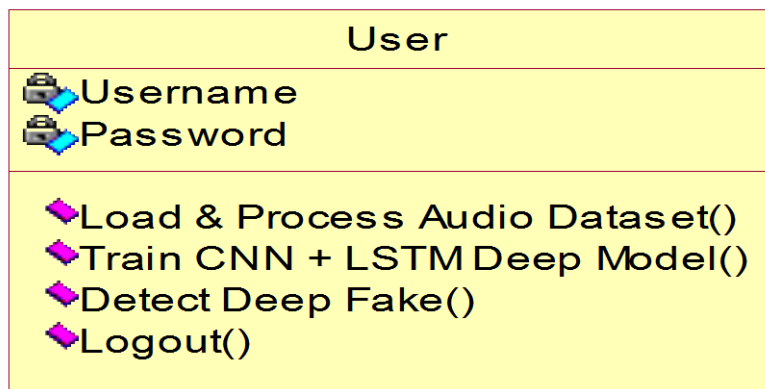


Fig.9.1 Data Flow Diagram

### 9.2. USECASE DIAGRAM:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.

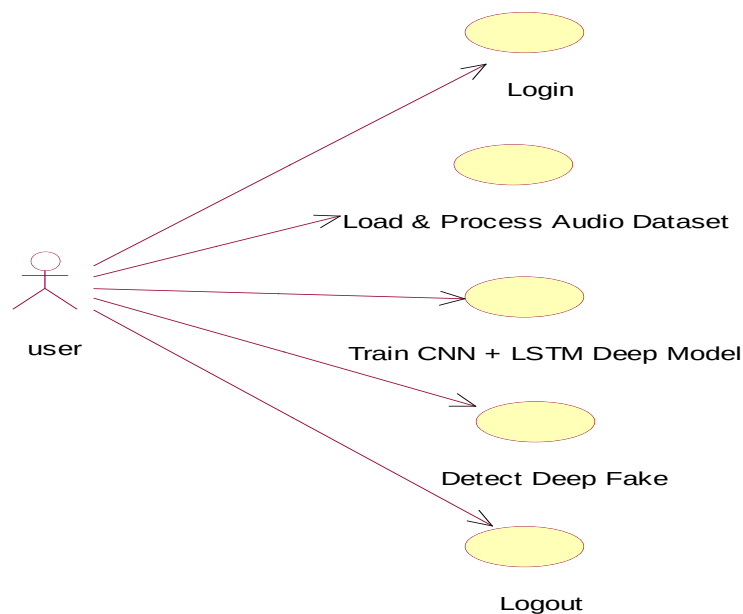


Fig.9.2 Use Case Diagram

### 9.3. SEQUENCE DIAGRAM:

A sequence diagram represents the interaction between different objects in the system. The important aspect of a sequence diagram is that it is time-ordered. This means that the exact sequence of the interactions between the objects is represented step by step. Different objects in the sequence diagram interact with each other by passing "messages".

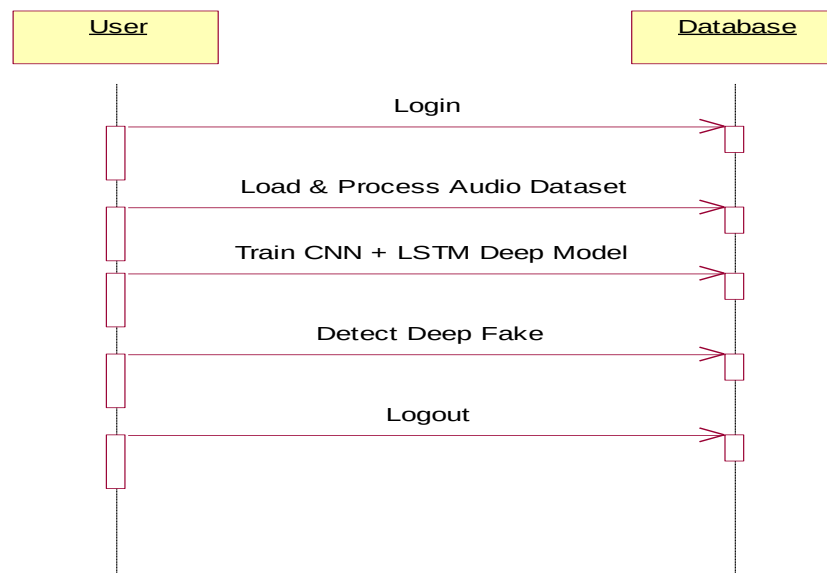


Fig.9.3 Sequence Diagram

#### 9.4. COLLABORATIVE DIAGRAM:

A collaboration diagram groups together the interactions between different objects. The interactions are listed as numbered interactions that help to trace the sequence of the interactions. The collaboration diagram helps to identify all the possible interactions that each object has with other objects.

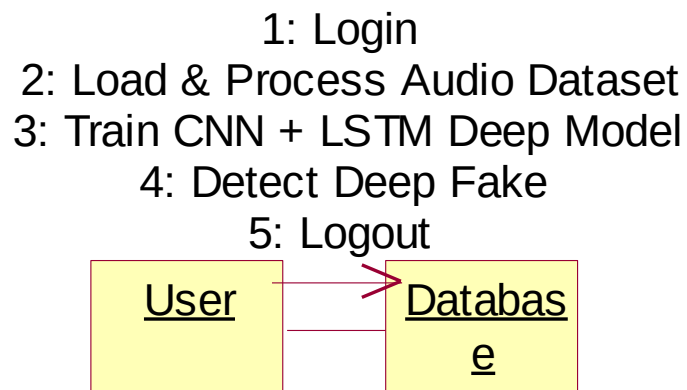


Fig.9.4 Collaborative Diagram

## 10.IMPLEMENTATION

### 10.1.SOURCE CODE:

**Settings.py :**

```
''''''
```

Django settings for DeepFake project.

Generated by 'django-admin startproject' using Django 2.0.

For more information on this file, see

<https://docs.djangoproject.com/en/2.0/topics/settings/>

For the full list of settings and their values, see

<https://docs.djangoproject.com/en/2.0/ref/settings/>

```
''''''
```

```
import os
```

```
# Build paths inside the project like this: os.path.join(BASE_DIR, ...)
```

```
BASE_DIR = os.path.dirname(os.path.dirname(os.path.abspath(__file__)))
```

```
SECRET_KEY = '8lzt+6b!5y9ik2_k4sv-^8z7itougw4=hiab5&o_1olfhkge&5'
```

```
# SECURITY WARNING: don't run with debug turned on in production!
```

```
DEBUG = True
```

```
ALLOWED_HOSTS = []
```



## Deep Fake Audio Detection using Deep Learning

# Application definition

```
INSTALLED_APPS = [  
  
    'django.contrib.admin',  
  
    'django.contrib.auth',  
  
    'django.contrib.contenttypes',  
  
    'django.contrib.sessions',  
  
    'django.contrib.messages',  
  
    'django.contrib.staticfiles',  
  
    'DeepFakeApp'  
]  
  
MIDDLEWARE = [  
  
    'django.middleware.security.SecurityMiddleware',  
  
    'django.contrib.sessions.middleware.SessionMiddleware',  
  
    'django.middleware.common.CommonMiddleware',  
  
    'django.middleware.csrf.CsrfViewMiddleware',  
  
    'django.contrib.auth.middleware.AuthenticationMiddleware',  
  
    'django.contrib.messages.middleware.MessageMiddleware',
```

## Deep Fake Audio Detection using Deep Learning

```
'django.middleware.clickjacking.XFrameOptionsMiddleware',  
  
]  
  
ROOT_URLCONF = 'DeepFake.urls'  
  
TEMPLATES = [  
  
    {  
  
        'BACKEND': 'django.template.backends.django.DjangoTemplates',  
  
        'DIRS': [],  
  
        'APP_DIRS': True,  
  
        'OPTIONS': {  
  
            'context_processors': [  
  
                'django.template.context_processors.debug',  
  
                'django.template.context_processors.request',  
  
                'django.contrib.auth.context_processors.auth',  
  
                'django.contrib.messages.context_processors.messages',  
  
            ],  
  
        },  
  
    ],  
  
]
```

## Deep Fake Audio Detection using Deep Learning

```
WSGI_APPLICATION = 'DeepFake.wsgi.application'

# Database

# https://docs.djangoproject.com/en/2.0/ref/settings/#databases

DATABASES = {

    'default': {

        'ENGINE': 'django.db.backends.sqlite3',

        'NAME': os.path.join(BASE_DIR, 'db.sqlite3'),

    }

}

# Password validation

# https://docs.djangoproject.com/en/2.0/ref/settings/#auth-password-validators

AUTH_PASSWORD_VALIDATORS = [

    {

        'NAME': 'django.contrib.auth.password_validation.UserAttributeSimilarityValidator',

    },

    {

        'NAME': 'django.contrib.auth.password_validation.MinimumLengthValidator',

    },

]
```

## Deep Fake Audio Detection using Deep Learning

```
{  
    'NAME': 'django.contrib.auth.password_validation.CommonPasswordValidator',  
},  
  
{  
    'NAME': 'django.contrib.auth.password_validation.NumericPasswordValidator',  
},  
]
```

LANGUAGE\_CODE = 'en-us'

TIME\_ZONE = 'UTC'

USE\_I18N = True

USE\_L10N = True

USE\_TZ = True

STATIC\_URL = '/static/'

### **Urls.py:**

```
"""DeepFake URL Configuration
```

The `urlpatterns` list routes URLs to views. For more information please see:

<https://docs.djangoproject.com/en/2.0/topics/http/urls/>

Examples:

### Function views

1. Add an import: `from my_app import views`
2. Add a URL to `urlpatterns`: `path("", views.home, name='home')`

### Class-based views

1. Add an import: `from other_app.views import Home`
2. Add a URL to `urlpatterns`: `path("", Home.as_view(), name='home')`

### Including another URLconf

1. Import the `include()` function: `from django.urls import include, path`
2. Add a URL to `urlpatterns`: `path('blog/', include('blog.urls'))`

```
.....
```

```
from django.contrib import admin
```

```
from django.urls import path, include
```

```
urlpatterns = [
```

```
    path('admin/', admin.site.urls),
```

```
    path("", include('DeepFakeApp.urls')),
```

```
]
```

```
wsgi.py:
```

```
.....
```

WSGI config for DeepFake project.

It exposes the WSGI callable as a module-level variable named ``application``.

For more information on this file, see

<https://docs.djangoproject.com/en/2.0/howto/deployment/wsgi/>

```
"""
```

```
import os
```

```
from django.core.wsgi import get_wsgi_application
```

```
os.environ.setdefault("DJANGO_SETTINGS_MODULE", "DeepFake.settings")
```

```
application = get_wsgi_application()
```

DeepFakeApp:

**urls.py:**

```
from django.urls import path
```

```
from . import views
```

```
urlpatterns = [path("index.html", views.index, name="index"),
```

```
                path("UserLogin.html", views.UserLogin, name="UserLogin"),
```

```
                path("UserLoginAction", views.UserLoginAction, name="UserLoginAction"),
```

```
                path("Predict.html", views.Predict, name="Predict"),
```

```
                path("PredictAction", views.PredictAction, name="PredictAction"),
```

## Deep Fake Audio Detection using Deep Learning

```
path("LoadDataset", views.LoadDataset, name="LoadDataset"),  
  
path("TrainModel", views.TrainModel, name="TrainModel"),  
  
]
```

### **views.py:**

```
from django.shortcuts import render  
  
from django.template import RequestContext  
  
from django.contrib import messages  
  
from django.http import HttpResponseRedirect  
  
from django.conf import settings  
  
import os  
  
import io  
  
import base64  
  
import matplotlib.pyplot as plt  
  
import pandas as pd  
  
import numpy as np  
  
from sklearn.model_selection import train_test_split  
  
from sklearn.metrics import accuracy_score  
  
from sklearn.metrics import f1_score
```

## Deep Fake Audio Detection using Deep Learning

```
from sklearn.metrics import precision_score

from sklearn.metrics import recall_score

import seaborn as sns

from sklearn.metrics import confusion_matrix

import os

import soundfile

import librosa

from sklearn.preprocessing import MinMaxScaler

from keras.utils.np_utils import to_categorical

from sklearn.model_selection import train_test_split

from keras.callbacks import ModelCheckpoint

import pickle

from keras.models import Sequential

from keras.layers import MaxPooling2D

from keras.layers import Dense, Dropout, Activation, Flatten, TimeDistributed, LSTM

from keras.layers import Conv2D

from keras.models import Sequential, load_model, Model

global username
```



## Deep Fake Audio Detection using Deep Learning

```
global X_train, X_test, y_train, y_test, X, Y

accuracy = []

precision = []

recall = []

fscore = []

#function to calculate all metrics

def calculateMetrics(algorithm, y_test, predict):

    a = (accuracy_score(y_test,predict)*100)

    p = (precision_score(y_test, predict,average='macro') * 100)

    r = (recall_score(y_test, predict,average='macro') * 100)

    f = (f1_score(y_test, predict,average='macro') * 100)

    a = round(a, 3)

    p = round(p, 3)

    r = round(r, 3)

    f = round(f, 3)

    accuracy.append(a)

    precision.append(p)

    recall.append(r)
```

```
fscore.append(f)

return algorithm

def extract_feature(file_name, mfcc, chroma, mel):

    with soundfile.SoundFile(file_name) as sound_file:

        X = sound_file.read(dtype="float32")

        sample_rate=sound_file.samplerate

        if chroma:

            stft=np.abs(librosa.stft(X))

            result=np.array([])

            if mfcc:

                mfccs=np.mean(librosa.feature.mfcc(y=X, sr=sample_rate, n_mfcc=40).T, axis=0)

                result=np.hstack((result, mfccs))

            if chroma:

                chroma=np.mean(librosa.feature.chroma_stft(S=stft, sr=sample_rate).T,axis=0)

                result=np.hstack((result, chroma))

            if mel:

                mel=np.mean(librosa.feature.melspectrogram(X, sr=sample_rate).T,axis=0)

                result=np.hstack((result, mel))
```

## Deep Fake Audio Detection using Deep Learning

```
sound_file.close()

return result

def values(filename, acc):

    f = open(filename, 'rb')

    train_values = pickle.load(f)

    f.close()

    accuracy_value = train_values[acc]

    return accuracy_value

X = np.load('model/X.npy')

Y = np.load('model/Y.npy')

X = np.reshape(X, (X.shape[0], (X.shape[1] * X.shape[2] * X.shape[3] * X.shape[4])))

scaler = MinMaxScaler((0,1))

X = scaler.fit_transform(X)

X = np.reshape(X, (X.shape[0], 1, 10, 6, 3))

indices = np.arange(X.shape[0])

np.random.shuffle(indices)

X = X[indices]

Y = Y[indices]
```

## Deep Fake Audio Detection using Deep Learning

```
Y = to_categorical(Y)

X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2) #split dataset into train and test

data = np.load("model/data.npy", allow_pickle=True)

X_train, X_test, y_train, y_test = data

cnn_lstm = Sequential()

#adding cnn layer with time distributed to extract spatial features

cnn_lstm.add(TimeDistributed(Conv2D(32, (3, 3), padding='same',activation = 'relu'), input_shape =
(X.shape[1], X.shape[2], X.shape[3], X.shape[4])))

#max layer will collect relevant features from CNN

cnn_lstm.add(TimeDistributed(MaxPooling2D((4, 4))))

#dropout will remove ir-relevant features

cnn_lstm.add(Dropout(0.5))

#adding another layer to further optimize features

cnn_lstm.add(TimeDistributed(Conv2D(64, (3, 3), padding='same',activation = 'relu')))

cnn_lstm.add(TimeDistributed(MaxPooling2D((1, 1))))

cnn_lstm.add(Dropout(0.5))

cnn_lstm.add(TimeDistributed(Conv2D(128, (3, 3), padding='same',activation = 'relu')))

cnn_lstm.add(TimeDistributed(MaxPooling2D((1, 1))))

cnn_lstm.add(Dropout(0.5))
```

## Deep Fake Audio Detection using Deep Learning

```
cnn_lstm.add(TimeDistributed(Conv2D(256, (2, 2), padding='same', activation = 'relu')))  
  
cnn_lstm.add(TimeDistributed(MaxPooling2D((1, 1))))  
  
cnn_lstm.add(Dropout(0.5))  
  
cnn_lstm.add(TimeDistributed(Flatten()))  
  
#adding LSTM layer to extract and optimzie temporal features  
  
cnn_lstm.add(LSTM(32))  
  
cnn_lstm.add(Dense(units = y_train.shape[1], activation = 'softmax'))  
  
#compiling, training and loading model  
  
cnn_lstm.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = ['accuracy'])  
  
if os.path.exists("model/cnn_lstm.hdf5") == False:  
  
    model_check_point = ModelCheckpoint(filepath='model/cnn_lstm.hdf5', verbose = 1, save_best_only = True)  
  
    hist = cnn_lstm.fit(X_train, y_train, batch_size = 32, epochs = 50, validation_data=(X_test, y_test),  
callbacks=[model_check_point], verbose=1)  
  
    f = open('model/cnn_lstm_history.pckl', 'wb')  
  
    pickle.dump(hist.history, f)  
  
    f.close()  
  
else:  
  
    cnn_lstm.load_weights("model/cnn_lstm.hdf5")  
  
#performing prediction on test data
```

## Deep Fake Audio Detection using Deep Learning

```
predict = cnn_lstm.predict(X_test)

predict = np.argmax(predict, axis=1)

y_test1 = np.argmax(y_test, axis=1)

#call funcction to calculate accuracy and other metrics

calculateMetrics("CNN + LSTM Algorithm", y_test1, predict)

conf_matrix = confusion_matrix(y_test1, predict)

def Predict(request):

    if request.method == 'GET':

        return render(request, 'Predict.html', {})

def PredictAction(request):

    if request.method == 'POST':

        global scaler

        class_label = ['Fake', 'Original']

        myfile = request.FILES['t1'].read()

        name = request.FILES['t1'].name

        if os.path.exists('DeepFakeApp/static/'+name):

            os.remove('DeepFakeApp/static/'+name)

        with open('DeepFakeApp/static/'+name, "wb") as file:
```

## Deep Fake Audio Detection using Deep Learning

```
file.write(myfile)

file.close()

cnn_lstm = load_model("model/cnn_lstm.hdf5")

mfcc = extract_feature("DeepFakeApp/static/"+name, mfcc=True, chroma=True, mel=True)

mfcc = np.reshape(mfcc, (10, 6, 3))

test = []

test.append([mfcc])

test = np.asarray(test)

test = np.reshape(test, (test.shape[0], (test.shape[1] * test.shape[2] * test.shape[3] * test.shape[4])))

test = scaler.transform(test)

test = np.reshape(test, (test.shape[0], 1, 10, 6, 3))

predict = cnn_lstm.predict(test)

predict = np.argmax(predict)

predict = class_label[predict]

context= {'data': '<font size="3" color="blue">Uploaded Audio Predicted As : '+predict+'</font>'}

return render(request, 'UserScreen.html', context)

def TrainModel(request):

    if request.method == 'GET':
```

## Deep Fake Audio Detection using Deep Learning

```
global X_train, X_test, y_train, y_test

global accuracy, precision, recall, fscore, conf_matrix

class_label = ['Fake', 'Original']

output='<table border=1 align=center width=100%><tr><th><font size="3" color="black">Algorithm
Name</th><th><font size="3" color="black">Accuracy</th>'

        output += '<th><font size="3" color="black">Precision</th><th><font size="3"
color="black">Recall</th><th><font size="3" color="black">FSCORE</th></tr>'

algorithms = ['CNN + LSTM Algorithm']

for i in range(len(algorithms)):

        output += '<td><font size="3" color="black">'+algorithms[i]+'</td><td><font size="3"
color="black">'+str(accuracy[i)+'</td><td><font size="3" color="black">'+str(precision[i)+'</td>'

        output += '<td><font size="3" color="black">'+str(recall[i)+'</td><td><font size="3"
color="black">'+str(fscore[i)+'</td></tr>'

output+= "</table></br>"

arch1_acc = values("model/cnn_lstm_history.pckl", "accuracy")

index = [1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 17, 18, 19, 20, 21, 22, 23, 24, 25, 26, 27, 28, 29,
30, 31, 32, 33, 34, 35, 36, 37,

38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 50]

figure, axis = plt.subplots(nrows=1, ncols=2,figsize=(10, 3))#display original and predicted segmented
image

axis[0].set_title("Confusion Matrix Prediction Graph")
```



## Deep Fake Audio Detection using Deep Learning

```
axis[1].set_title("CNN Training Accuracy Graph")

    ax = sns.heatmap(conf_matrix, xticklabels = class_label, yticklabels = class_label, annot = True,
cmap="viridis" ,fmt ="g", ax=axis[0]);

    ax.set_ylim([0,len(class_label)])

axis[1].plot(index, arch1_acc, color="green")

axis[1].legend(['Training Accuracy'], loc='lower right')

buf = io.BytesIO()

plt.savefig(buf, format='png', bbox_inches='tight')

#plt.close()

img_b64 = base64.b64encode(buf.getvalue()).decode()

plt.clf()

plt.cla()

context= {'data':output, 'img': img_b64}

return render(request, 'UserScreen.html', context)

def LoadDataset(request):

    if request.method == 'GET':

        global X_train, X_test, y_train, y_test, X, Y

        class_label = ['Fake', 'Original']

        output = '<font size="3" color="black">Deep Fake Audio Dataset Loaded</font><br/>'
```

## Deep Fake Audio Detection using Deep Learning

```
output += '<font size="3" color="blue">Total Audio MFCC Features Extracted from each Audio =  
180</font><br/>'
```

```
output += '<font size="3" color="blue">Different Class Labels found in Dataset =  
' + str(class_label) + '</font><br/><br/>'
```

```
output += '<font size="3" color="black">Dataset Train & Test Split details</font><br/>'
```

```
output += '<font size="3" color="blue">80% dataset MFCC features used to train Algorithm =  
' + str(X_train.shape[0] * 180) + '</font><br/>'
```

```
output += '<font size="3" color="blue">20% dataset MFCC features used to test Algorithm =  
' + str(X_test.shape[0] * 180) + '</font><br/>'
```

```
context= {'data':output}
```

```
return render(request, 'UserScreen.html', context)
```

```
def UserLoginAction(request):
```

```
    global username
```

```
    if request.method == 'POST':
```

```
        global username
```

```
        username = request.POST.get('t1', False)
```

```
        password = request.POST.get('t2', False)
```

```
        if username == 'admin' and password == 'admin':
```

```
            context= {'data':'Welcome '+username}
```

```
            return render(request, "UserScreen.html", context)
```

## Deep Fake Audio Detection using Deep Learning

else:

context= {'data':'Invalid username'}

return render(request, 'UserLogin.html', context)

def UserLogin(request):

if request.method == 'GET':

return render(request, 'UserLogin.html', {})

def index(request):

if request.method == 'GET':

return render(request, 'index.html', {})

**manage.py:**

#!/usr/bin/env python

import os

import sys

if \_\_name\_\_ == "\_\_main\_\_":

os.environ.setdefault("DJANGO\_SETTINGS\_MODULE", "DeepFake.settings")

try:

from django.core.management import execute\_from\_command\_line

except ImportError as exc:

```
raise ImportError(

    "Couldn't import Django. Are you sure it's installed and "

    "available on your PYTHONPATH environment variable? Did you "

    "forget to activate a virtual environment?"

) from exc

execute_from_command_line(sys.argv)
```

### **test.py:**

```
import soundfile

import librosa

import os

import numpy as np

from sklearn.preprocessing import MinMaxScaler

from keras.utils.np_utils import to_categorical

from sklearn.model_selection import train_test_split

from keras.callbacks import ModelCheckpoint

import pickle

from keras.models import Sequential

from sklearn.metrics import accuracy_score
```

## Deep Fake Audio Detection using Deep Learning

```
from keras.layers import MaxPooling2D

from keras.layers import Dense, Dropout, Activation, Flatten, TimeDistributed, LSTM

from keras.layers import Conv2D

from keras.models import Sequential, load_model, Model

def extract_feature(file_name, mfcc, chroma, mel):

    with soundfile.SoundFile(file_name) as sound_file:

        X = sound_file.read(dtype="float32")

        sample_rate=sound_file.samplerate

        if chroma:

            stft=np.abs(librosa.stft(X))

            result=np.array([])

            if mfcc:

                mfccs=np.mean(librosa.feature.mfcc(y=X, sr=sample_rate, n_mfcc=40).T, axis=0)

                result=np.hstack((result, mfccs))

            if chroma:

                chroma=np.mean(librosa.feature.chroma_stft(S=stft, sr=sample_rate).T,axis=0)

                result=np.hstack((result, chroma))

            if mel:
```

## Deep Fake Audio Detection using Deep Learning

```
mel=np.mean(librosa.feature.melspectrogram(X, sr=sample_rate).T,axis=0)

result=np.hstack((result, mel))

sound_file.close()

return result

path = "Dataset"

'''

X = []

Y = []

for root, dirs, directory in os.walk(path):

    for j in range(len(directory)):

        name = os.path.basename(root)

        mfcc = extract_feature(root+"/"+directory[j], mfcc=True, chroma=True, mel=True)

        label = 0

        if name == 'Original':

            label = 1

        mfcc = np.reshape(mfcc, (10, 6, 3))

        X.append([mfcc])

        Y.append(label)
```

## Deep Fake Audio Detection using Deep Learning

```
print(str(mfcc.shape)+" "+str(label)+" "+name)

X = np.asarray(X)

Y = np.asarray(Y)

np.save("model/X", X)

np.save("model/Y", Y)

'''

X = np.load("model/X.npy")

Y = np.load("model/Y.npy")

print(X.shape)

X = np.reshape(X, (X.shape[0], (X.shape[1] * X.shape[2] * X.shape[3] * X.shape[4])))

print(X.shape)

scaler = MinMaxScaler((0,1))

X = scaler.fit_transform(X)

X = np.reshape(X, (X.shape[0], 1, 10, 6, 3))

indices = np.arange(X.shape[0])

np.random.shuffle(indices)

X = X[indices]
```

## Deep Fake Audio Detection using Deep Learning

```
Y = Y[indices]
```

```
Y = to_categorical(Y)
```

```
X_train, X_test, y_train, y_test = train_test_split(X, Y, test_size=0.2) #split dataset into train and test
```

```
data = np.load("model/data.npy", allow_pickle=True)
```

```
X_train, X_test, y_train, y_test = data
```

```
lstm_model = Sequential()
```

```
lstm_model.add(TimeDistributed(Conv2D(32, (3, 3), padding='same',activation = 'relu'), input_shape =  
(X.shape[1], X.shape[2], X.shape[3], X.shape[4])))
```

```
lstm_model.add(TimeDistributed(MaxPooling2D((4, 4))))
```

```
lstm_model.add(Dropout(0.5))
```

```
lstm_model.add(TimeDistributed(Conv2D(64, (3, 3), padding='same',activation = 'relu')))
```

```
lstm_model.add(TimeDistributed(MaxPooling2D((1, 1))))
```

```
lstm_model.add(Dropout(0.5))
```

```
lstm_model.add(TimeDistributed(Conv2D(128, (3, 3), padding='same',activation = 'relu')))
```

```
lstm_model.add(TimeDistributed(MaxPooling2D((1, 1))))
```

```
lstm_model.add(Dropout(0.5))
```

```
lstm_model.add(TimeDistributed(Conv2D(256, (2, 2), padding='same',activation = 'relu')))
```

```
lstm_model.add(TimeDistributed(MaxPooling2D((1, 1))))
```

```
lstm_model.add(Dropout(0.5))
```



## Deep Fake Audio Detection using Deep Learning

```
lstm_model.add(TimeDistributed(Flatten()))

lstm_model.add(LSTM(32))

lstm_model.add(Dense(units = y_train.shape[1], activation = 'softmax'))

lstm_model.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = ['accuracy'])

if os.path.exists("model/cnn_lstm.hdf5") == False:

    model_check_point = ModelCheckpoint(filepath='model/cnn_lstm.hdf5', verbose = 1, save_best_only = True)

    hist = lstm_model.fit(X_train, y_train, batch_size = 32, epochs = 50, validation_data=(X_test, y_test),
callbacks=[model_check_point], verbose=1)

    f = open('model/cnn_lstm_history.pckl', 'wb')

    pickle.dump(hist.history, f)

    f.close()

else:

    lstm_model.load_weights("model/cnn_lstm.hdf5")

predict = lstm_model.predict(X_test)

predict = np.argmax(predict, axis=1)

y_test1 = np.argmax(y_test, axis=1)

acc = accuracy_score(y_test1, predict)

print(acc)
```

## 11.SCREENSHOTS

In propose work we are utilizing combination of CNN and LSTM to detect deep fake audio. CNN (Convolutional Neural Network) and LSTM (Long Short-Term Memory) are combined for deepfake detection because CNN excels at extracting spatial features from audio frames, while LSTM analyses temporal patterns and inconsistencies over time, leading to a more comprehensive and accurate detection method.

CNNs for Spatial Feature Extraction:

CNNs are highly effective at identifying visual patterns and features within individual frames of a video, such as facial textures, lighting, and other spatial details.

LSTMs for Temporal Analysis:

LSTMs are designed to process sequential data, making them well-suited for analysing the temporal relationships between video frames. They can identify inconsistencies or unnatural movements that might be indicative of a deep fake.

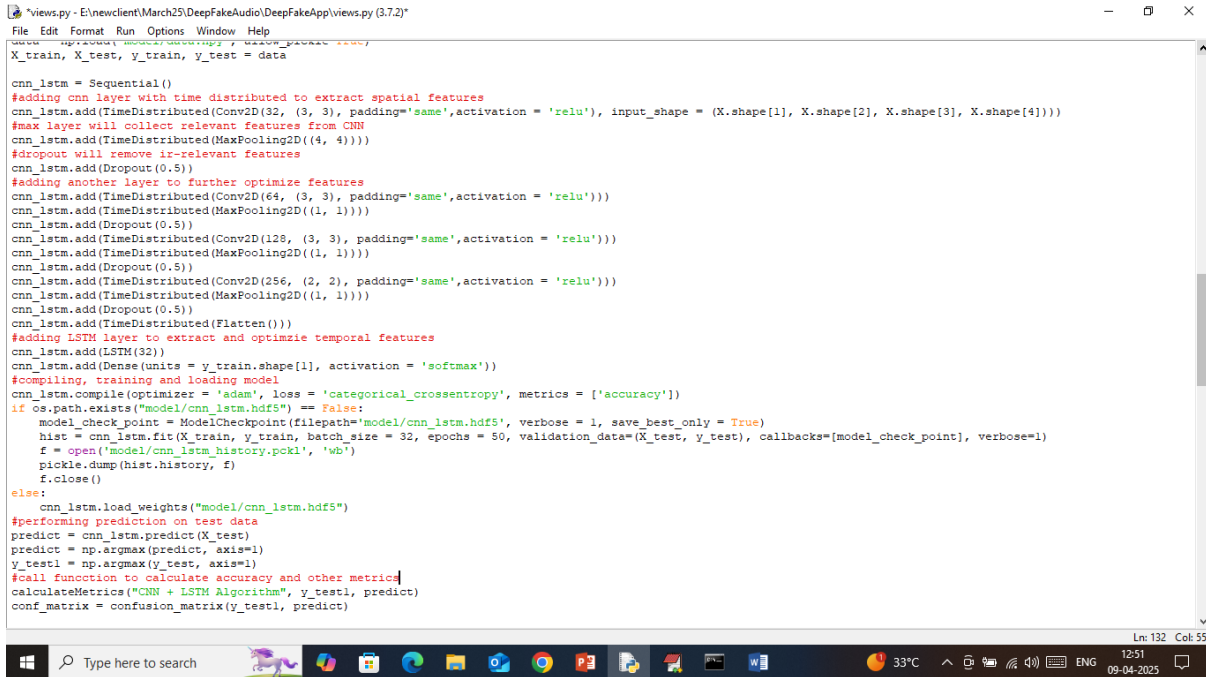
To train above algorithms we have used deep fake audio dataset from KAGGLE repository which can be download from below URL

<https://www.kaggle.com/datasets/f0rtaza/fake-audio/data>

Each audio from above dataset is processed and extracted MFCC features and then perform shuffling and normalization to prepare training array. All processed training array will be split into train and test where application using 80% data for training and 20% for testing.

80% training features will be input to deep fake CNN + LSTM algorithm to train a model and this model will be applied on 20% test data to calculate prediction accuracy. In below screen showing CNN + LSTM layers along with time distributed layer to capture inconsistency over time.

## Deep Fake Audio Detection using Deep Learning

A screenshot of a Windows File Explorer window showing a Python script named 'views.py' located at 'E:\newclient\March25\DeepFakeAudio\DeepFakeApp\views.py (3.7.2)'. The script is a Jupyter Notebook file. The code defines a CNN + LSTM model for audio detection. It includes comments in red text explaining the purpose of each layer and step. The model is trained on a dataset of audio features (MFCC) and evaluated on a test set. The script uses TensorFlow/Keras for the CNN and LSTM layers, and NumPy for data manipulation. The output of the script is a confusion matrix and accuracy metrics.

```
"views.py - E:\newclient\March25\DeepFakeAudio\DeepFakeApp\views.py (3.7.2)"
File Edit Format Run Options Window Help
data = load_data('data', 'data', 'data', 'data', 'data')
X_train, X_test, y_train, y_test = data

cnn_lstm = Sequential()
#adding cnn layer with time distributed to extract spatial features
cnn_lstm.add(TimeDistributed(Conv2D(32, (3, 3), padding='same', activation = 'relu'), input_shape = (X.shape[1], X.shape[2], X.shape[3], X.shape[4])))
#max layer will collect relevant features from CNN
cnn_lstm.add(TimeDistributed(MaxPooling2D((4, 4))))
#dropout will remove ir-relevant features
cnn_lstm.add(Dropout(0.5))
#adding another layer to further optimize features
cnn_lstm.add(TimeDistributed(Conv2D(64, (3, 3), padding='same', activation = 'relu')))
cnn_lstm.add(TimeDistributed(MaxPooling2D((1, 1))))
cnn_lstm.add(Dropout(0.5))
cnn_lstm.add(TimeDistributed(Conv2D(128, (3, 3), padding='same', activation = 'relu')))
cnn_lstm.add(TimeDistributed(MaxPooling2D((1, 1))))
cnn_lstm.add(Dropout(0.5))
cnn_lstm.add(TimeDistributed(Conv2D(256, (2, 2), padding='same', activation = 'relu')))
cnn_lstm.add(TimeDistributed(MaxPooling2D((1, 1))))
cnn_lstm.add(Dropout(0.5))
cnn_lstm.add(TimeDistributed(Flatten()))
#adding LSTM layer to extract and optimize temporal features
cnn_lstm.add(LSTM(32))
cnn_lstm.add(Dense(units = y_train.shape[1], activation = 'softmax'))
#compiling, training and loading model
cnn_lstm.compile(optimizer = 'adam', loss = 'categorical_crossentropy', metrics = ['accuracy'])
if os.path.exists("model/cnn_lstm.hdf5") == False:
    model_check_point = ModelCheckpoint(filepath='model/cnn_lstm.hdf5', verbose = 1, save_best_only = True)
    hist = cnn_lstm.fit(X_train, y_train, batch_size = 32, epochs = 50, validation_data=(X_test, y_test), callbacks=[model_check_point], verbose=1)
    f = open("model/cnn_lstm_history.pkl", 'wb')
    pickle.dump(hist.history, f)
    f.close()
else:
    cnn_lstm.load_weights("model/cnn_lstm.hdf5")
#performing prediction on test data
predict = cnn_lstm.predict(X_test)
predict = np.argmax(predict, axis=1)
y_test1 = np.argmax(y_test, axis=1)
#call function to calculate accuracy and other metrics
calculateMetrics("CNN + LSTM Algorithm", y_test1, predict)
conf_matrix = confusion_matrix(y_test1, predict)
```

Fig.11.1

In above screen read red colour comments to know about CNN + LSTM deep fake audio detection model.

To implement this project we have designed following modules

- 1) **User Login:** user can login to system using username and password as 'admin and admin'
- 2) **Load & Process Audio Dataset:** using this model will load and normalize all dataset audio MFCC features and then split into train and test where application using 80% images for training and 20% for testing
- 3) **Train CNN + LSTM Deep Model:** 80% training MFCC features will be input to deep learning CNN + LSTM algorithm to trained a model and this model will be applied on 20% test images to calculate prediction accuracy
- 4) **Detect Deep Fake:** using this module user can upload test audio and then application will extract MFCC features and then input to CNN + LSTM model to predict weather audio is original or Deep Fake.

Install python 3.7.2 and then install all packages given in requirements.txt file and then double click on 'run.bat' file to start python server and then will get below page

## Deep Fake Audio Detection using Deep Learning

```
C:\Windows\system32\cmd.exe
C:\Users\Admin\AppData\Local\Programs\Python\Python37\lib\site-packages\tensorflow\python\framework\dtypes.py:517: FutureWarning: Passing (type, 1) or '1type' as a synonym of type is deprecated; in a future version of numpy, it will be understood as (type, (1,)) / '(1,)type'.
  _np_qint8 = np.dtype(("qint8", np.uint8, 1))
C:\Users\Admin\AppData\Local\Programs\Python\Python37\lib\site-packages\tensorflow\python\framework\dtypes.py:518: FutureWarning: Passing (type, 1) or '1type' as a synonym of type is deprecated; in a future version of numpy, it will be understood as (type, (1,)) / '(1,)type'.
  _np_qint16 = np.dtype(("qint16", np.int16, 1))
C:\Users\Admin\AppData\Local\Programs\Python\Python37\lib\site-packages\tensorflow\python\framework\dtypes.py:519: FutureWarning: Passing (type, 1) or '1type' as a synonym of type is deprecated; in a future version of numpy, it will be understood as (type, (1,)) / '(1,)type'.
  _np_qint32 = np.dtype(("qint32", np.int32, 1))
C:\Users\Admin\AppData\Local\Programs\Python\Python37\lib\site-packages\tensorflow\python\framework\dtypes.py:520: FutureWarning: Passing (type, 1) or '1type' as a synonym of type is deprecated; in a future version of numpy, it will be understood as (type, (1,)) / '(1,)type'.
  _np_resource = np.dtype(("resource", np.ubyte, 1))
WARNING:tensorflow:From C:\Users\Admin\AppData\Local\Programs\Python\Python37\lib\site-packages\keras\backend\tensorflow_backend.py:4070: The name tf.nn.max_pool is deprecated. Please use tf.nn.max_pool2d instead.
WARNING:tensorflow:From C:\Users\Admin\AppData\Local\Programs\Python\Python37\lib\site-packages\keras\backend\tensorflow_backend.py:422: The name tf.global_variables is deprecated. Please use tf.compat.v1.global_variables instead.

System check identified no issues (0 silenced).

You have 14 unapplied migration(s). Your project may not work properly until you apply the migrations for app(s): admin, auth, contenttypes, sessions.
Run 'python manage.py migrate' to apply them.
April 09, 2025 - 12:52:36
Django version 2.0, using settings 'DeepFake.settings'
Starting development server at http://127.0.0.1:8000/
Quit the server with CTRL-BREAK.
```

Fig.11.2

In above screen python server started and now open browser and enter URL as <http://127.0.0.1:8000/index.html> and then press enter key to get below page

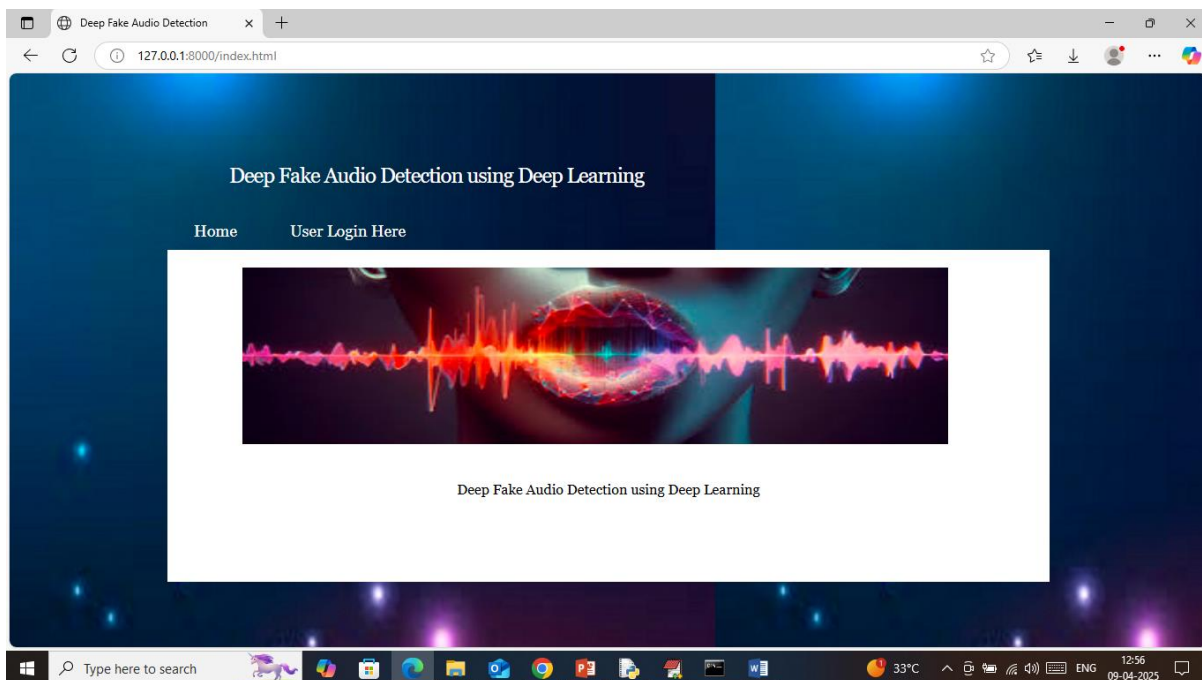


Fig.11.3

## Deep Fake Audio Detection using Deep Learning

In above screen click on 'User Login' link to get below page

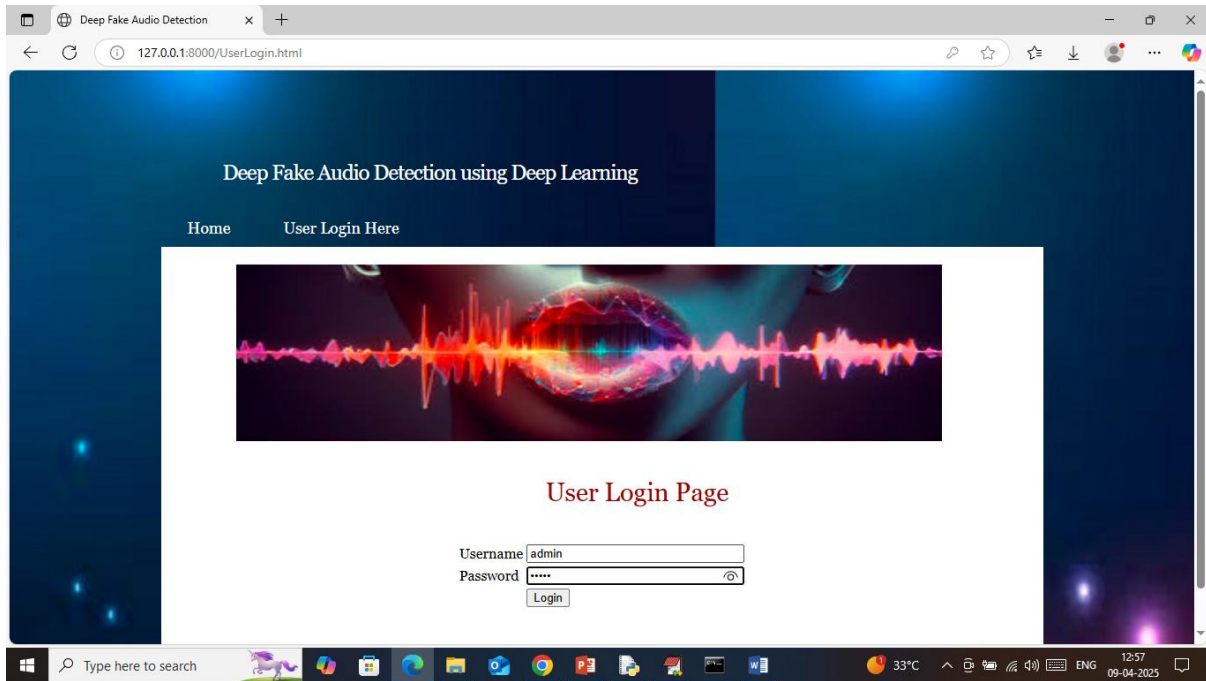


Fig.11.4

In above screen user is login and after login will get below page

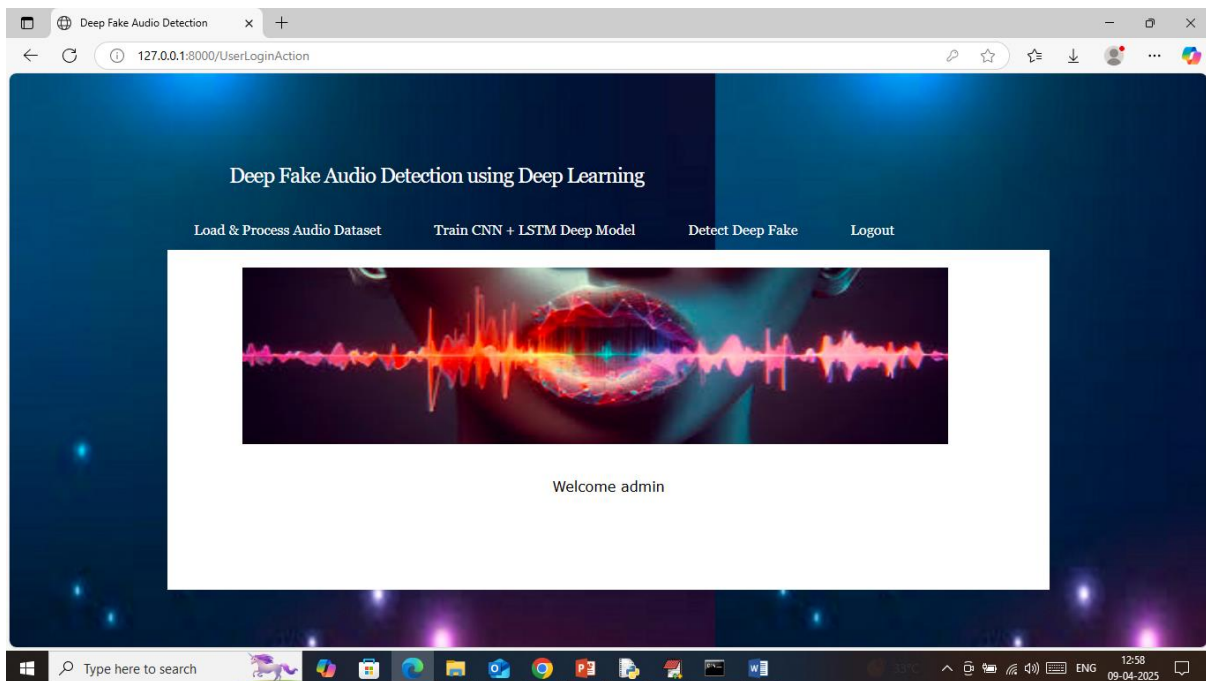


Fig.11.5

## Deep Fake Audio Detection using Deep Learning

In above screen user can click on 'Load & Process Audio Dataset' link to load dataset and then will get below page

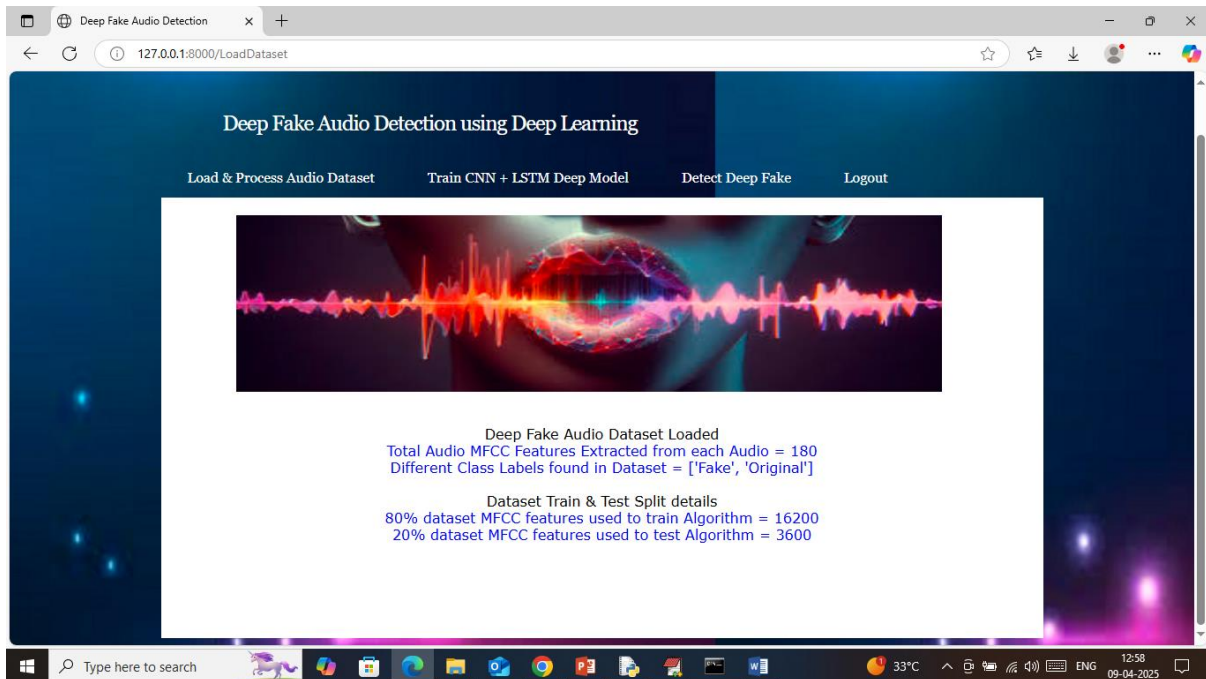


Fig.11.6

In above screen can see number of audio files loaded and processed from dataset and then can see train and test size. Now click on 'Train CNN + LSTM Deep Model' link to train al CNN algorithm and then will get below page

## Deep Fake Audio Detection using Deep Learning

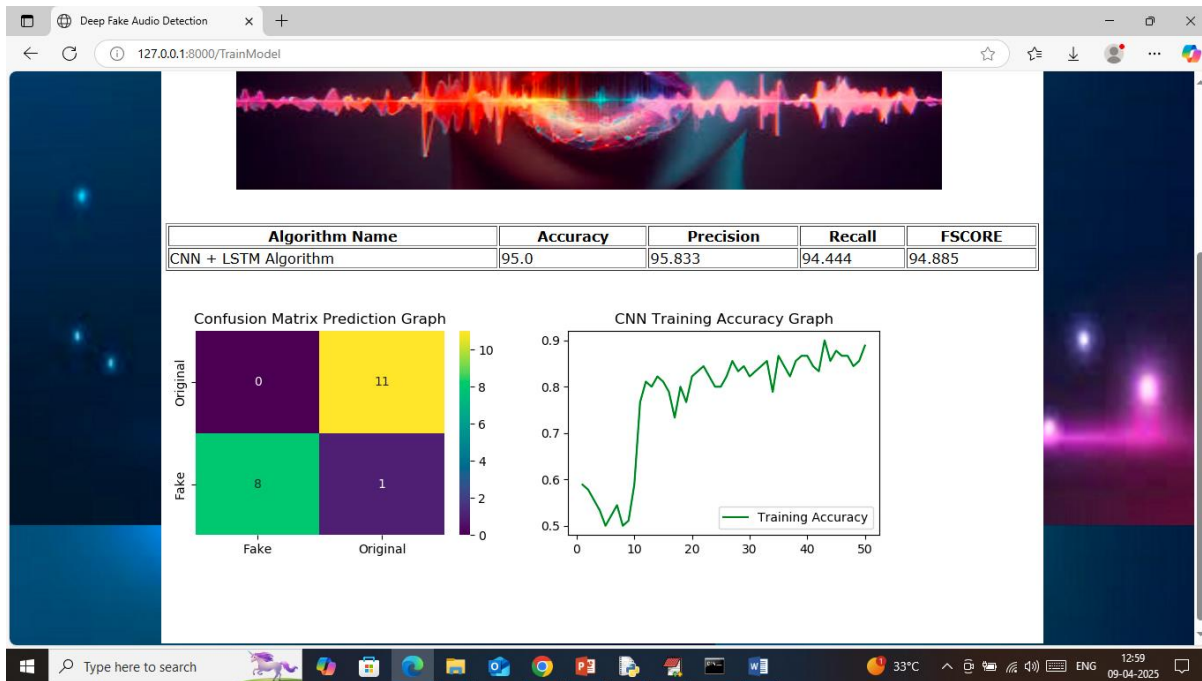


Fig.11.7

In above screen in table format can see accuracy, precision, recall, FSCORE of CNN + LSTM algorithm. In above screen can see CNN can detect deep fake audio with an accuracy of 95%. In confusion matrix graph x-axis represents Predicted Labels and y-axis represents true labels and then yellow and green boxes in diagonal represents correct prediction count and remaining blue boxes represents incorrect prediction count which are very few. In second graph can see training accuracy of CNN where x-axis represents 'Number of training epochs' and y-axis represents 'accuracy' and can see with each increasing epoch accuracy got increased and reached closer to 1. Now click on 'Detect Deep Fake' link to get below page



## Deep Fake Audio Detection using Deep Learning

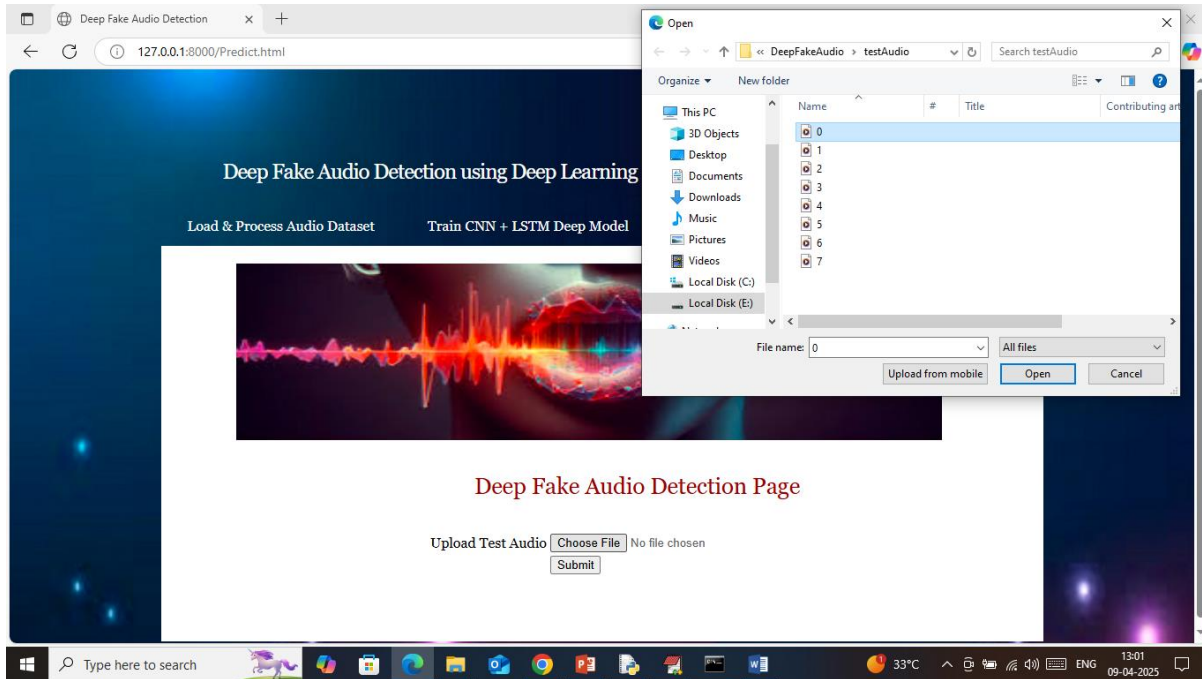


Fig.11.8

In above screen select and upload test audio file and then click on ‘Open and Submit’ button to get below output

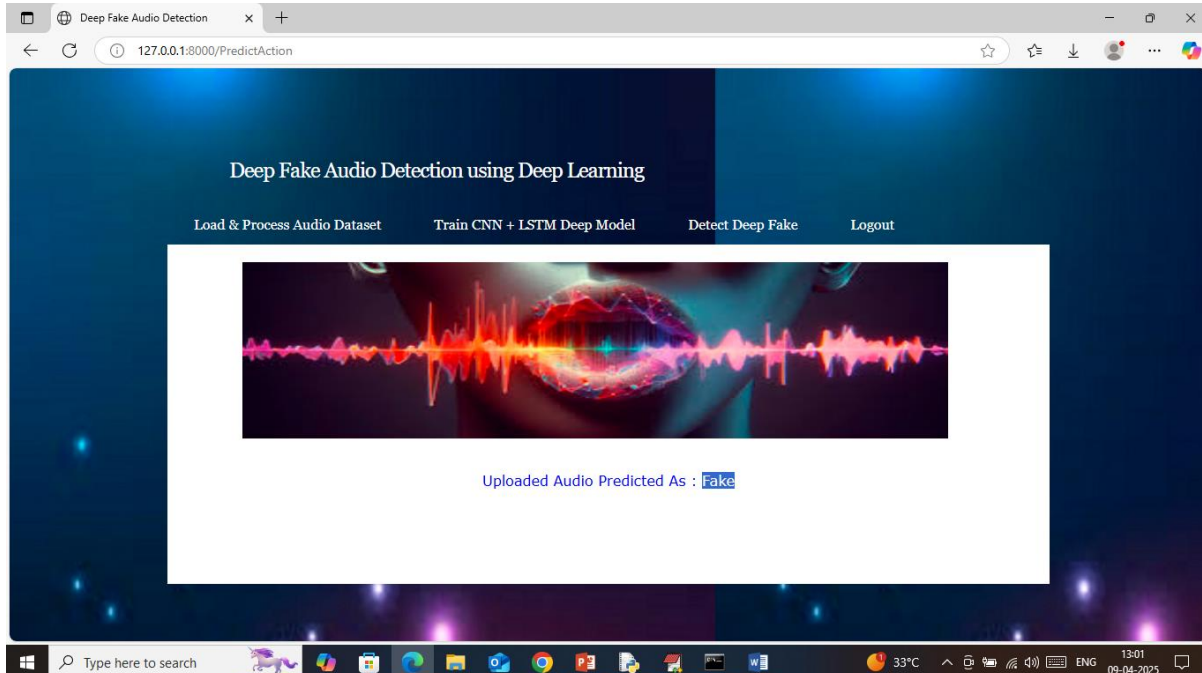


Fig.11.9

In above screen uploaded audio detected as “Fake” and similarly you can upload and test other videos



## Deep Fake Audio Detection using Deep Learning

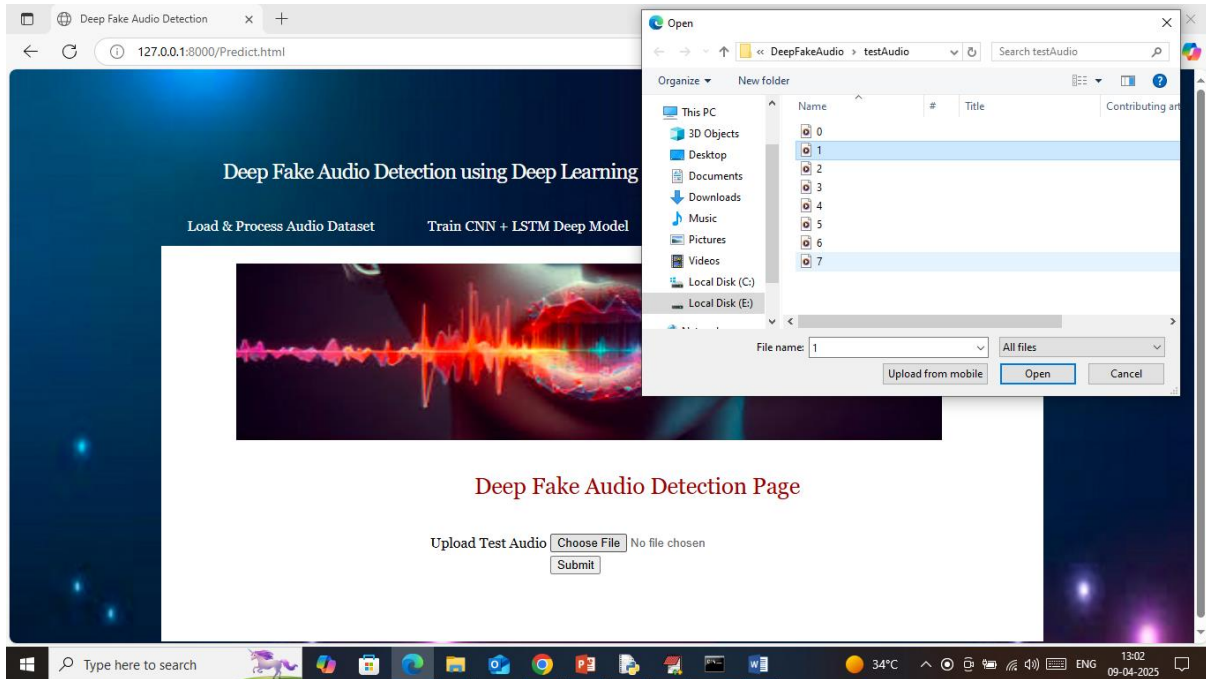


Fig.11.10

In above screen uploading another audio file and below is the output

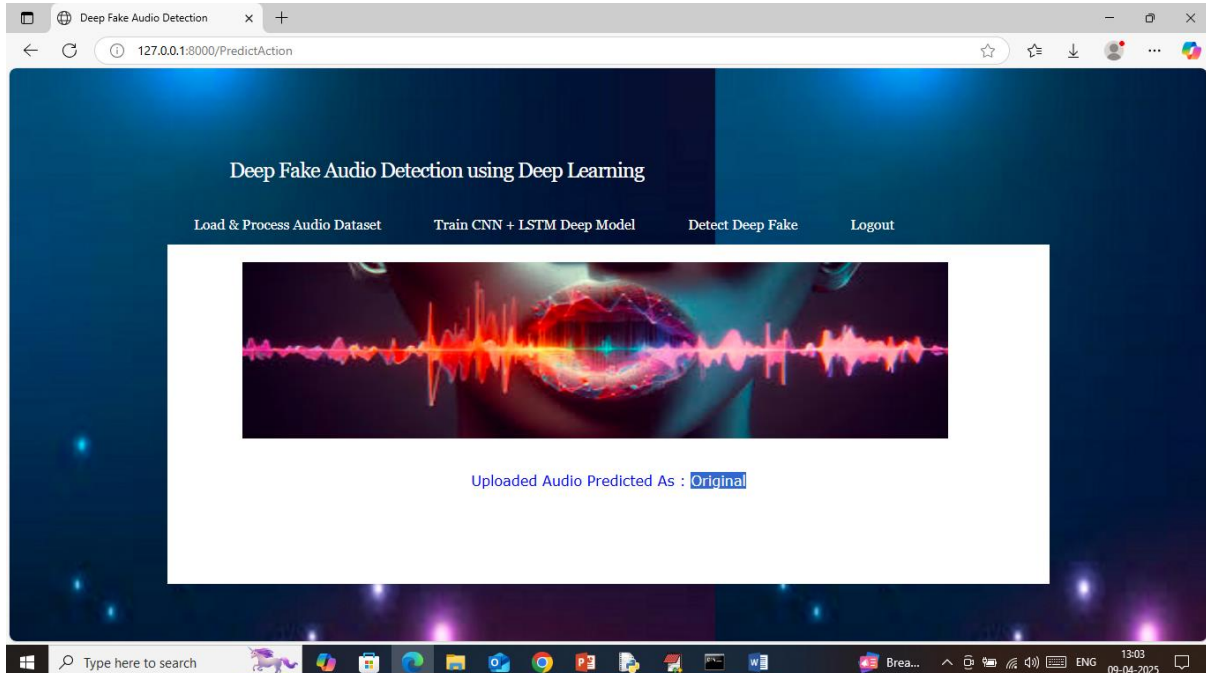


Fig.11.11

## 12.TESTING

Testing is an important step in our project to make sure that the system works correctly and gives accurate results. The main goal of testing is to identify errors and ensure that the deepfake audio detection system performs as expected.

In this project, different types of testing were carried out to verify the functionality, performance, and reliability of the system.

### 12.1.Types of Testing

#### 12.1.1. Unit Testing:

Unit testing is done to check individual components of the system. Each module, such as audio preprocessing, feature extraction, and model prediction, was tested separately to ensure it works properly.

For example, we verified whether:

Audio files are loaded correctly

MFCC features are extracted properly

Model gives output without errors

#### 12.1.2. Integration Testing:

Integration testing is used to check whether all modules work together correctly. After combining modules like dataset processing, model training, and prediction, we tested the system as a whole.

This ensures:

Smooth data flow between modules

No errors when modules interact with each other

#### 12.1.3. Functional Testing:

Functional testing checks whether the system behaves according to the requirements. It ensures that all functions are working as expected.

Some of the functional checks include:

Uploading an audio file

Processing the audio

Predicting whether the audio is real or fake

Displaying the result correctly

### **12.1.4. System Testing:**

System testing is performed on the complete system to verify overall performance. It ensures that the entire application works correctly in a real environment.

This includes:

Running the full pipeline from input to output

Checking system response time

Ensuring correct predictions

### **12.1.5. User Acceptance Testing (UAT):**

User Acceptance Testing is done to ensure that the system meets user expectations. The system was tested by users to confirm that it is easy to use and provides correct results.

## **12.2. Test Cases**

### **Test Case 1: Login Function**

Function: User Login

Expected Result: System should validate user credentials

Actual Result: User successfully logged in with valid credentials

Status: Passed

### **Test Case 2: Dataset Processing**

Function: Load and process dataset

Expected Result: Audio files should be loaded and features extracted

## Deep Fake Audio Detection using Deep Learning

Actual Result: Dataset processed successfully

Status: Passed

### **Test Case 3:** Model Training

Function: Train CNN + LSTM model

Expected Result: Model should train and show performance metrics

Actual Result: Model trained successfully with high accuracy

Status: Passed

### **Test Case 4:** Deepfake Detection

Function: Upload and detect audio

Expected Result: System should classify audio as real or fake

Actual Result: Correct classification displayed

Status: Passed

**Test Results:** All the test cases were executed successfully. The system performed well in all scenarios without major errors. The model achieved high accuracy and correctly classified most of the audio samples.

## 13.CONCLUSION

This project explores and implements a deep learning-based approach to detecting deepfake audio using spectrogram analysis and convolutional neural networks (CNNs). By converting audio signals into time-frequency representations (e.g., Mel-spectrograms), the model learns to identify minute inconsistencies that often arise during synthetic voice generation, such as unnatural harmonics, background inconsistencies, and temporal artifacts.

Experimental results demonstrate that deep learning models, particularly CNNs and LSTMs, can achieve high accuracy in distinguishing between real and fake audio samples across multiple datasets and voice synthesis techniques. The trained model shows robustness in generalization, achieving reliable performance even when tested on previously unseen deepfake techniques. This confirms that deep learning offers a viable defense mechanism in the fight against malicious audio forgery.

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